Niklas Reinhardt

Mail ◇ LinkedIn ◇ Website

Education

Heidelberg University — Heidelberg, Germany
PhD, topic: Statistical Learning in Infinite Dimensions (expected completion 12/2027) — 01/2024 – present

Heidelberg University — Heidelberg, Germany
M. Sc. Mathematics and M. Sc. Physics, final grade 1.0 in both (best possible grade) — 03/2021 – 10/2023

Karlsruhe Institute of Technology (KIT) — Karlsruhe, Germany
B. Sc. Mathematics and B. Sc. Physics, final grade 1.1 in both (best: 1.0) — 10/2017 – 02/2021

Otto-Hahn-Gymnasium — Nagold, Germany
Abitur, final grade 1.0 (top of class); advanced subjects mathematics, chemistry and physics — 2010 – 2017

Research Experience

PhD Research — Heidelberg University
Statistical learning in infinite dimensions — 01/2024 – present

- **Expressivity of Bayesian Neural Networks** (since 03/2026): analyzing approximation and expressivity properties of Bayesian neural networks in function space; studying, in theory and in code, whether randomizing only a few weights suffices, with applications to diffusion models and SPDEs.
- **Bayesian Operator Learning** (2025 – 2026): developed a Bayesian operator-learning framework for parameter-dependent PDEs; proved minimax learning rates for a class of smooth operators and implemented the method in Python for the diffusion equation.
- **Statistical Learning Theory for Operator Learning** (2024): analyzed sample complexity and generalization of neural operators; showed that they learn the diffusion equation's solution operator at algebraic rates, avoiding the curse of dimensionality.

Master's Theses — Heidelberg University
Mathematics and Physics — 02/2022 – 10/2023

- **Open-loop stable solutions of a two-legged robot** (Mathematics): multi-phase optimal control.
- **Modified Gravity Theories with Screening Effects in KFT** (Physics): numerics in Python/C++ (JCAP).

Publications and Preprints

- Bayesian Operator Learning: Posterior Existence and Convergence of Point Estimates for Gaussian Priors. Preprint 10/2026
- Statistical learning theory for neural operators. *JMLR* 06/2026
- Kinetic field theory: effects of modified gravity theories with screening mechanisms on non-linear cosmic density fluctuations. *JCAP* 03/2025

Prizes and Awards

- Mobility and travel grants: heiDOCS 2.0, HGS MathComp, Oberwolfach Leibniz Graduate Students
- Abitur prizes in mathematics, physics and German 2017
- Several first and second prizes in the Bundeswettbewerb and Landeswettbewerb Mathematik (national and state mathematics competitions) 2013, 2014

Teaching and Scientific Activities

- **Head Assistant** (2024 – present): organizing courses and supervising 150+ students and 5 tutors ("Theory of Deep Learning" 2026, 2024; "Einführung in die Numerik" 2026; "High-Dimensional Numerics" 2025).
- **Student Mentoring** (2024 – present): supervision of several bachelor's and master's theses.
- **Talks and Posters**: Oberwolfach, SIAM CSE (2027), SciCADE, Swiss Numerics Day (2026), ENUMATH (2025), several workshops.
- **Peer Reviewer**: reviewed papers for ENUMATH proceedings, *Bernoulli* and *Neural Networks*.

Research Interests and Skills

- **Research interests**: statistical learning theory, operator learning, Bayesian inference, uncertainty quantification
- **Programming**: Python (NumPy, pandas, scikit-learn, PyTorch), C++, Git, LaTeX